# Exploratory Data Analysis (EDA) Report  Cryptocurrency Volatility Prediction

## 1.Introduction:

Exploratory Data Analysis (EDA) is a critical step in any data science pipeline, aimed at understanding the structure, quality, and underlying patterns within the dataset before model building. In this study, EDA was conducted on historical cryptocurrency market data to analyze price behavior, liquidity characteristics, and their relationship with short-term volatility. The insights obtained from EDA guided feature selection and validated the feasibility of predicting 7-day rolling volatility using machine learning techniques.

### 2. Dataset Overview and Data Quality

The dataset consists of historical records of multiple cryptocurrencies, including price and market-related attributes such as **open, high, low, close prices, trading volume, and market capitalization**.

### 2.1 Data Structure

- The dataset contains multiple cryptocurrencies identified by crypto_name.
- Time-based columns (date, timestamp) were converted into datetime format.
- Data was sorted by cryptocurrency name and date to maintain temporal consistency, which is essential for rolling window calculations.

### 2.2 Data Cleaning

- **Daily returns** were computed using percentage change in closing prices.
- **Rolling volatility (7-day standard deviation of returns)** was calculated as the target variable.
- Rolling computations and percentage changes introduced missing values.
- Infinite values caused by division in the liquidity ratio were handled.
- Rows containing NaN or infinite values were removed to ensure compatibility with machine learning models.

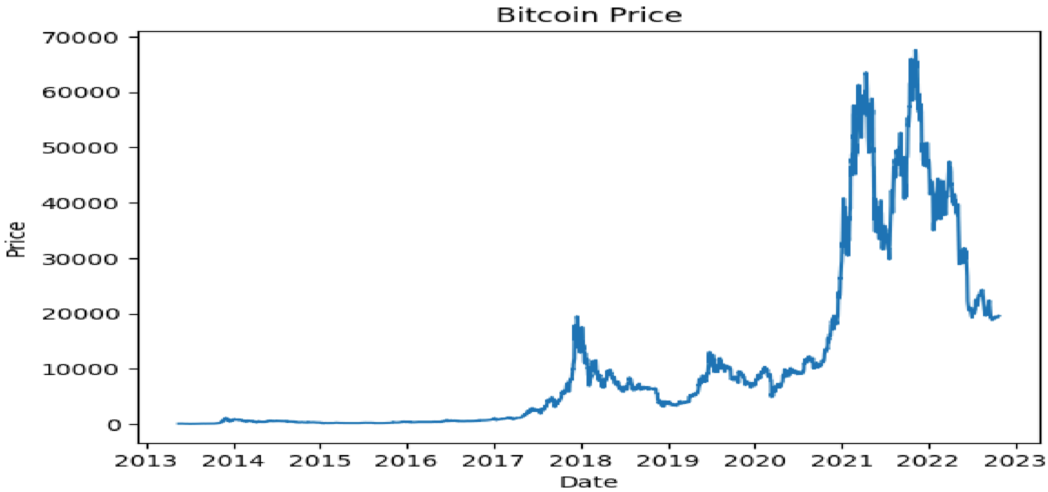
This preprocessing ensured that the dataset used for modeling was clean, stable, and statistically meaningful.

# 3. Time-Series Trend Analysis

To better understand market behavior, **Bitcoin** was selected for visualization due to its high liquidity and relatively stable trading behavior compared to other cryptocurrencies.

## 3.1 Price Movement Analysis

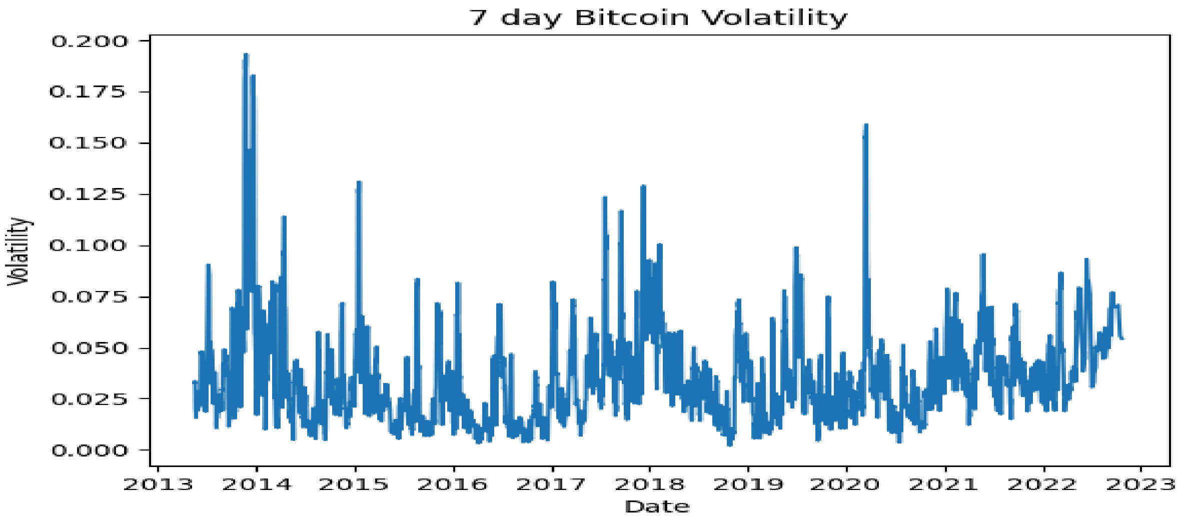
**Figure 1: Bitcoin Closing Price Over Time**




The price trend shows long-term growth phases interspersed with sharp corrections. These fluctuations highlight the inherently volatile nature of cryptocurrency markets and justify the need for volatility modeling.

## 3.2 Volatility Behavior

**Figure 2: 7-Day Rolling Volatility of Bitcoin**

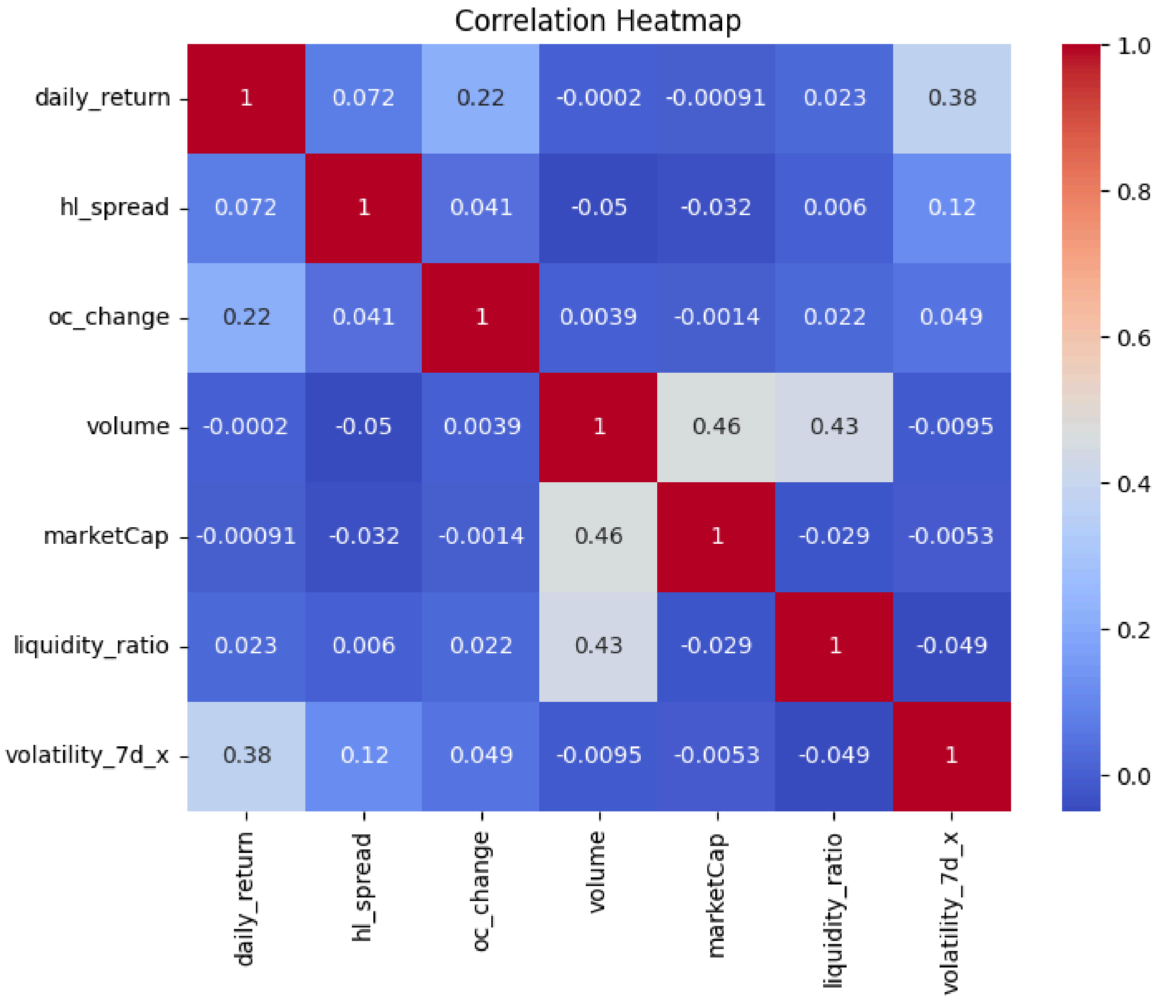


The volatility plot reveals **volatility clustering**, where periods of high volatility tend to persist. This behavior aligns with well-known financial market theories and indicates that past volatility contains predictive information about future risk.

# 4. Correlation Analysis

To quantify relationships between features and the target variable, a correlation matrix was generated.

**Figure 3: Correlation Heatmap of Key Features**




## Key Observations:

- **High-Low Spread (hl_spread)** shows a strong positive correlation with volatility, indicating that large intraday price ranges are reliable indicators of future risk.
- **Daily Returns** exhibit a noticeable relationship with volatility, as extreme price changes often precede periods of heightened uncertainty.
- **Liquidity Ratio** demonstrates meaningful correlation, suggesting that assets with higher trading activity relative to market capitalization tend to experience greater volatility.

- Raw **volume and market capitalization** alone show weaker direct correlations but become more informative when combined as a ratio.

# 5. Feature Behavior and Market Interpretation

## 5.1 Intraday Price Instability

Features such as **open-close change (oc_change)** and **high-low spread** capture short-term market stress. Large values in these features reflect strong buying or selling pressure within a single trading day.

## 5.2 Liquidity Effects

The **liquidity ratio (volume / market cap)** helps distinguish actively traded cryptocurrencies from illiquid ones. Coins with unusually high liquidity ratios often show sharper price movements, contributing to increased volatility.

## 5.3 Trend Smoothing

Moving averages (7-day and 14-day) were included to capture short-term and medium-term price trends while smoothing noisy fluctuations. These features help the model contextualize current prices within recent market history.

# 6. EDA Summary and Insights for Modeling

The exploratory analysis confirms that cryptocurrency volatility is **not random** and exhibits identifiable patterns linked to price spreads, returns, and liquidity conditions.

## Key Takeaways:

- Volatility shows **temporal dependency** and clustering behavior.
- Intraday price range and liquidity-based features are strong predictors of short-term volatility.
- Feature engineering significantly improves the model's ability to capture market dynamics.
- The EDA validates the use of a **Random Forest Regressor**, as it can effectively model nonlinear relationships between engineered features and volatility.

The insights derived from EDA formed a strong foundation for feature selection, preprocessing strategy, and model choice in the subsequent modeling phase.

# Pipeline Architecture of Cryptocurrency Volatility Prediction System

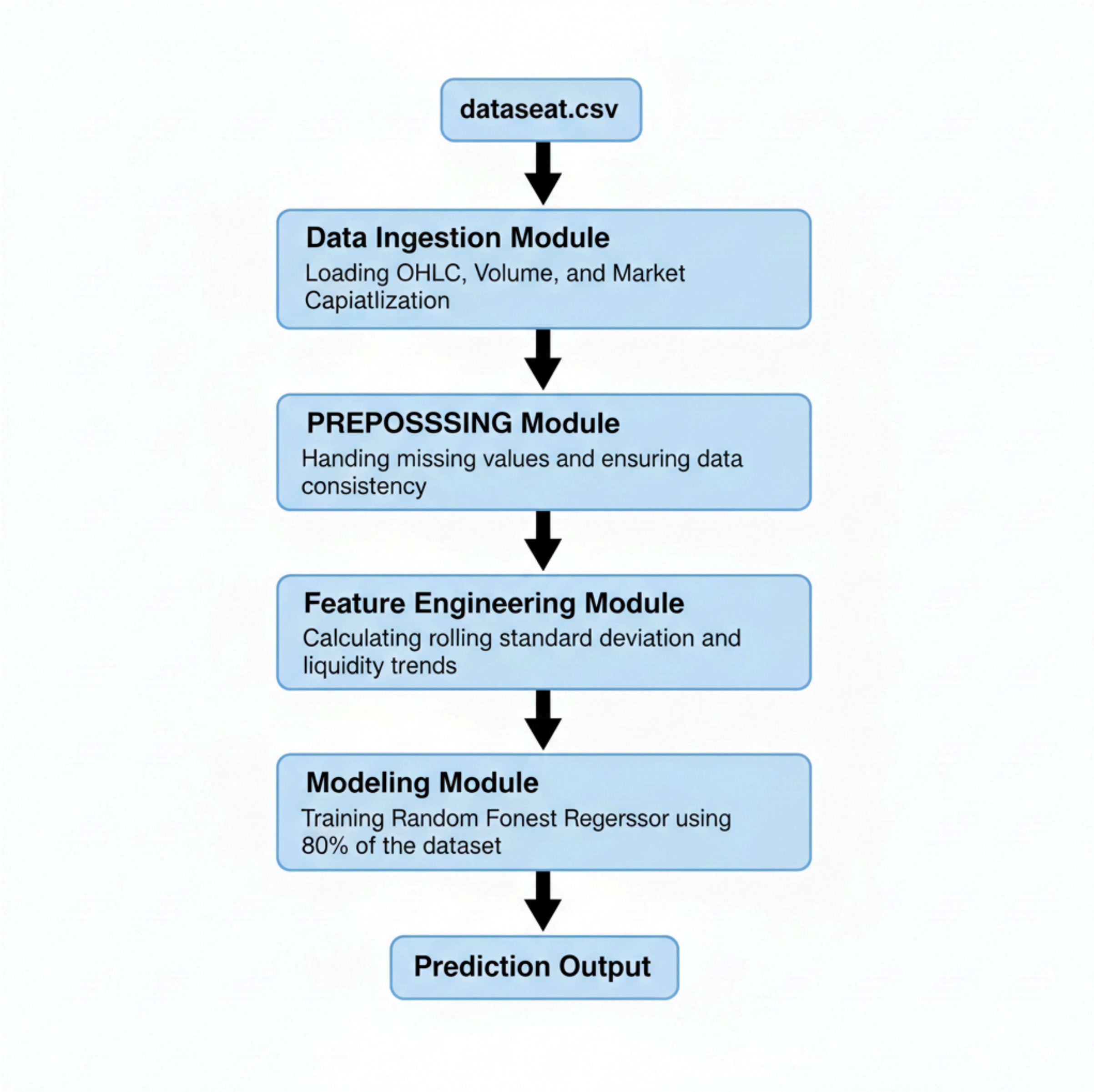



**Figure 4: Pipeline Architecture of Cryptocurrency Volatility Prediction System**

The above figure illustrates the end-to-end pipeline architecture of the cryptocurrency volatility prediction system. The pipeline begins with data ingestion from the dataset, followed by preprocessing and feature engineering stages. The processed data is then used to train a Random Forest regression model, and the final stage produces volatility predictions along with performance evaluation.